TARUNSAILESH VANGIPURAPU

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PROFESSIONAL SUMMARY

Senior AI Data Pipeline Engineer with 4+ years of experience designing, scaling, and operationalizing high-performance data pipelines and production-ready AI workflows that power global data-driven initiatives. Proven expertise engineering sophisticated streaming architectures and end-to-end AI data infrastructure — from real-time ingestion and transformation to BI reporting — with deep hands-on experience in Medallion Architecture, Delta Lake, PySpark, and distributed compute across Azure and AWS. Demonstrated ability to build the backbone of next-generation AI systems by delivering clean, governed, and scalable datasets that enable intelligent decision-making across complex, global operations. Certified in Azure AI Engineering, Fabric Data Engineering, and Databricks Data Engineering — ready to shape the future of intelligence at scale.

PROFESSIONAL EXPERIENCE

Unilever | Data Engineer **Jan 2024 – Present**

Redmond, WA | Project: Logistics Supply Chain Management

- Architected production-ready, end-to-end AI data pipelines using Azure Data Factory and Databricks to centralize high-volume logistics and supply chain data into a Medallion Architecture (Bronze → Silver → Gold), establishing a single source of truth for global shipping, warehousing, reverse logistics, and delivery modules — driving operational excellence and enabling downstream AI and analytics workloads at scale.
- Engineered and maintained the Logistics Foundation Model — a robust, scalable data model implementing strict schema enforcement and data validation logic to ensure high data integrity across transport performance tracking, inventory management, and carrier planning metrics, directly supporting production-ready AI workflow consumption.
- Developed high-performance PySpark and SQL transformations to process large-scale, complex supply chain telemetry, applying distributed computing best practices and data storage principles to ensure accessibility, reliability, and integrity of datasets consumed by AI, BI, and analytics teams.
- Designed and managed data quality monitoring (DQM) frameworks and audit processes, including automated anomaly detection, failure alerting, and pipeline observability across the supply chain data ecosystem — maintaining SLA compliance for global data-driven initiatives.
- Built and scaled streaming data integration modules enabling real-time ingestion from diverse logistics sources — warehousing systems, transportation platforms, and vendor portals — leveraging AWS Kinesis and Firehose streaming architectures alongside batch workflows, using modular, reusable Python code aligned with software engineering best practices.
- Collaborated with cross-functional stakeholders across carrier planning, inventory, and operations to translate logistics KPIs into interactive Power BI dashboards using advanced DAX measures, delivering real-time visibility into contract compliance, shipping efficiency, and supply chain performance.
- Implemented automated CI/CD pipelines using Azure DevOps to streamline deployment of ADF pipelines and Databricks notebooks across Dev, QA, and Prod environments — reducing manual deployment errors by 30% and accelerating production-ready AI workflow releases.
- Optimized Azure Databricks resource utilization by implementing auto-scaling clusters, Delta Lake Z-Ordering, and intelligent partitioning strategies — achieving a 25% reduction in cloud compute costs while sustaining high throughput for daily supply chain batch and streaming loads.
- Enhanced data governance and compliance by implementing Unity Catalog within Databricks for fine-grained access control and end-to-end data lineage tracking, ensuring adherence to global data privacy regulations across sensitive logistics and vendor datasets.
- Integrated automated monitoring and alerting for AI data pipelines within Azure, proactively identifying and resolving throughput bottlenecks in delivery and shipping data streams to uphold SLA compliance and operational excellence across the global data platform.
- Designed and operationalized ML-ready feature datasets by standardizing data transformation logic and enriching Gold-layer outputs with derived logistics attributes — enabling data science teams to accelerate model training cycles and reduce feature engineering overhead by an estimated 35%.
- Spearheaded cross-team data integration efforts by onboarding 5+ new logistics data sources into the centralized pipeline ecosystem, defining ingestion schemas, mapping transformation rules, and validating data contracts — expanding the platform's coverage and supporting broader AI-driven supply chain intelligence initiatives.

Wipro Pvt Ltd | Data / Software Engineer **May 2021 – Jan 2023**

Hyderabad, India | Project: ICM Analytics / Enterprise Data Platforms

- Led end-to-end migration of 60+ legacy data workloads (Databricks notebooks) to Azure Synapse Analytics, re-architecting ETL pipelines into scalable, production-ready AI data infrastructure — eliminating premium cluster dependency and achieving 30% infrastructure cost reduction while maintaining SLA compliance.
- Optimized PySpark-based AI data pipeline workflows by refactoring expensive transformations, improving partitioning strategies and execution plans — delivering a 40% reduction in job runtime and enhanced pipeline stability for enterprise reporting and analytics workloads.

- Designed and configured dedicated Apache Spark pools in Azure Synapse and EMR-equivalent distributed compute environments, tuning executor memory, dynamic allocation, node sizing, and Spark configurations to deliver high-performance, cost-efficient streaming and batch processing architectures.
- Engineered and executed migration of legacy analytics workloads to Azure Synapse Analytics, reducing pipeline execution times from 28 minutes to 10 minutes through distributed computing optimization and intelligent pipeline re-architecture — directly accelerating data-driven decision cycles.
- Implemented secure data access patterns using managed identities and controlled mount paths to Azure Data Lake Storage (ADLS Gen2), strengthening data governance and environment isolation across Dev/Test/Prod; integrated non-relational data stores including document and key-value databases to support flexible, scalable AI data access patterns.
- Authored advanced KQL queries and engineered critical analytical functions within Azure Data Explorer (Kusto) to power telemetry analytics for operational reporting, post-incident analysis, and real-time intelligence workflows.
- Implemented comprehensive data quality monitoring across all production workspaces, establishing automated failure detection systems and authoring troubleshooting guides (TSGs) to ensure pipeline reliability and operational support readiness.
- Designed scalable data models in Azure Synapse to support high-performance analytical queries, building governed datasets that data scientists and analysts leveraged to generate actionable AI-driven insights for executive dashboards and operational reporting.
- Developed reusable Python utility libraries and parameterized pipeline templates to standardize data ingestion and transformation patterns across enterprise workloads — reducing development effort for new pipeline onboarding by approximately 30% and enforcing consistency across distributed engineering teams.
- Collaborated with data architects and ML engineers to define and deliver curated analytical datasets optimized for model training and experimentation, ensuring data freshness, schema stability, and lineage transparency across all AI-consumed data products on the enterprise platform.

TECHNICAL SKILLS

AI & ML Pipeline Infrastructure: Production-Ready AI Workflows, AI Data Pipeline Design, Streaming Architecture, Real-Time Ingestion, Feature Engineering Support, ML Dataset Preparation

Cloud Platforms: Azure (ADF, Databricks, Synapse Analytics, ADLS Gen2, Azure DevOps, Unity Catalog, Azure Monitor), AWS (Kinesis, Firehose, EMR, S3)

Streaming & Distributed Computing: Apache Spark, PySpark, AWS Kinesis, Firehose, Auto-Scaling Clusters, Dynamic Allocation, Distributed Compute Optimization

Data Pipeline & Orchestration: Azure Data Factory, Databricks Workflows, CI/CD Pipeline Automation, Medallion Architecture (Bronze/Silver/Gold), Delta Lake, Z-Ordering, Partitioning

Languages & Query: Python, PySpark, SQL, KQL (Kusto Query Language), DAX

Data Storage & Governance: Delta Lake, ADLS Gen2, Azure Synapse, Unity Catalog, Data Lineage Tracking, Schema Enforcement, Data Quality Monitoring (DQM)

BI & Visualization: Power BI, Advanced DAX, Executive Dashboards, Operational Reporting

DevOps & MLOps: Azure DevOps, CI/CD Automation, Environment Isolation (Dev/QA/Prod), Automated Alerting & Monitoring

Non-Relational Stores: Document Databases, Key-Value Databases, Flexible Data Access Patterns

EDUCATION

Master of Science in Information Systems Jan 2023 – Dec 2024

University of North Texas — Denton, TX

Relevant Coursework: Data Structures & Algorithms, DBMS, Distributed Systems, System Design, Computer Architecture, Operating Systems

Bachelor of Technology in Electronics & Communication Engineering Jun 2017 – May 2021

Gitam University — Hyderabad, India

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02) — Amazon Web Services
- Azure AI Engineer Associate (AI-102) — Microsoft
- Fabric Data Engineer Associate (DP-700) — Microsoft
- Data Engineer Professional — Databricks